

Transition Path Methods for Heterogeneous Agent Models

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Abstract

This document provides the mathematical foundations behind the transition path solver implemented in `hetmacro.transition`. We cover three methods for computing perfect foresight transition dynamics after an MIT shock: (i) backward forward shooting with damped iteration, (ii) root finding via Broyden’s method over capital, and (iii) root finding over interest rates. We then cast the Aiyagari transition path as a coordination problem in the sense of Angeletos and Lian (2016), showing how the nature of strategic interaction (substitutability vs. complementarity) governs both equilibrium uniqueness and the convergence of the numerical methods.

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1 Setup: MIT Shocks and Transition Dynamics

Consider a heterogeneous agent economy in steady state at $t = 0$. At $t = 1$, an unexpected (MIT) shock hits the economy. After the shock is realized, agents have perfect foresight over the entire future path of the economy. We seek to compute the transition path from the initial steady state back to the terminal steady state.

1.1 The Household Problem Along the Transition

Let a denote asset holdings and z idiosyncratic productivity. Along the transition, the household problem at date t is:

$$V_t(a, z) = \max_{c, a'} \left\{ u(c) + \beta \sum_{z'} \Pi(z' | z) V_{t+1}(a', z') \right\} \quad (1)$$

subject to the budget constraint:

$$c + a' = (1 + r_t)a + y_t(z), \quad a' \geq \underline{a} \quad (2)$$

where r_t is the interest rate at date t , $y_t(z) = w_t z$ is labor income, and Π is the Markov transition matrix for the idiosyncratic state. The key difference from the stationary problem is that the value function, prices, and policies are all time varying.

Terminal condition. At $t = T$ (the terminal date), we impose that the economy has returned to its terminal steady state:

$$V_T(a, z) = V^{\text{ss}}(a, z) \quad (3)$$

This pins down the backward recursion. The horizon T must be chosen large enough so that this boundary condition is innocuous.

1.2 Distribution Dynamics

Let $D_t(a, z)$ denote the joint distribution of assets and productivity at date t . Given a policy function $a'_t(a, z)$ and the Markov transition matrix Π , the distribution evolves forward as:

$$D_{t+1}(a', z') = \sum_z \Pi(z' | z) \sum_a \mathbf{1}\{a'_t(a, z) = a'\} D_t(a, z) \quad (4)$$

In practice, when policies do not land exactly on grid points, we use lottery weights (linear interpolation on the asset grid) to distribute mass across adjacent grid points.

Initial condition. At $t = 0$, the distribution is the initial steady state distribution:

$$D_0(a, z) = D^{\text{ss}}(a, z) \quad (5)$$

1.3 Market Clearing

In the Aiyagari model, the aggregate capital stock equals mean asset holdings:

$$K_t = \sum_{a,z} a \cdot D_t(a, z) \quad (6)$$

Given aggregate capital K_t and the exogenous shock path $\{Z_t\}_{t=0}^{T-1}$, prices are determined by firm optimality:

$$r_t = Z_t \alpha K_t^{\alpha-1} - \delta \quad (7)$$

$$w_t = Z_t (1 - \alpha) K_t^\alpha \quad (8)$$

The transition path problem is to find a capital path $\{K_t\}_{t=0}^{T-1}$ such that market clearing (6) holds at every date, given that household policies and distributions are consistent with the prices implied by this capital path.

1.4 Timing Convention

The timing follows the standard discrete time convention:

- $t = 0$: shock hits; agents re-optimize immediately.
- $D_0 = D^{\text{ss}}$: the pre-shock steady state distribution (assets are predetermined).
- $K_t = \sum_{a,z} a \cdot D_t(a, z)$: capital at time t equals mean assets that agents *bring into* period t .
- $a'_t(a, z)$: savings chosen at t , which become assets at $t + 1$.
- Backward iteration: $t = T - 1, T - 2, \dots, 0$.
- Forward iteration: $t = 0, 1, \dots, T - 1$.

2 Method 1: Backward Forward Shooting

The shooting method solves the transition path by iterating on the aggregate capital path. The idea is simple: guess a capital path, compute the implied household behavior, check whether the resulting distribution is consistent with the guessed capital, and update.

Algorithm 1 (Backward Forward Shooting with Damped Iteration).

Inputs: Initial steady state $(\bar{K}, \bar{r}, D^{\text{ss}}, V_a^{\text{ss}})$, terminal steady state, shock path $\{Z_t\}_{t=0}^{T-1}$, damping parameter $\lambda \in (0, 1)$, tolerance $\varepsilon > 0$.

Initialize: Set $K_t^{(0)} = \bar{K}$ for all t (or use linear interpolation between initial and terminal steady state capital).

Step 1. Compute prices. For each $t = 0, \dots, T - 1$:

$$r_t^{(n)} = Z_t \alpha \left(K_t^{(n)}\right)^{\alpha-1} - \delta, \quad w_t^{(n)} = Z_t (1 - \alpha) \left(K_t^{(n)}\right)^\alpha$$

Step 2. Backward iteration. Starting from the terminal marginal value V_a^{ss} at $t = T$, iterate backward using the Endogenous Grid Method (EGM):

$$\text{For } t = T - 1, T - 2, \dots, 0 : \quad V_{a,t}, a'_t, c_t = \text{EGM} \left(V_{a,t+1}, \Pi, \mathcal{A}, y_t, r_t, \beta, \sigma^{-1} \right)$$

where EGM denotes a single step of the endogenous grid method, which exploits the Euler equation:

$$u'(c_t(a, z)) = \beta(1 + r_{t+1}) \sum_{z'} \Pi(z' | z) u'(c_{t+1}(a'_t(a, z), z')) \quad (9)$$

This produces a sequence of policy functions $\{a'_t, c_t\}_{t=0}^{T-1}$.

Step 3. Forward iteration. Starting from $D_0 = D^{ss}$, iterate the distribution forward using the time varying policies:

$$\text{For } t = 0, 1, \dots, T - 1 : \quad D_{t+1} = \mathcal{T}(D_t, a'_t, \Pi)$$

where $\mathcal{T}$ denotes the forward operator that distributes mass using lottery weights on the asset grid. At each date, compute the implied capital:

$$\hat{K}_t = \sum_{a,z} a \cdot D_t(a, z) \quad (10)$$

Step 4. Convergence check. Compute the error:

$$e^{(n)} = \max_{0 \leq t \leq T-1} \left| \hat{K}_t - K_t^{(n)} \right|$$

If $e^{(n)} < \varepsilon$, stop and return $\hat{K}$ as the equilibrium path.

Step 5. Damped update. Update the capital path guess with damping:

$$K_t^{(n+1)} = \lambda K_t^{(n)} + (1 - \lambda) \hat{K}_t \quad (11)$$

Return to Step 1.

2.1 The Role of Damping

Without damping ($\lambda = 0$), the algorithm sets the next guess equal to the implied capital path. This often leads to oscillations: if the guess is too high, the implied path overshoots in the other direction, and so forth. Damping stabilizes convergence by taking a convex combination of the old guess and the new implied path.

The parameter λ controls the trade off between stability and speed:

- λ close to 1: very stable but slow convergence (small steps).
- λ close to 0: fast but potentially unstable (large steps).
- A good default is $\lambda = 2/3$, which corresponds to a step size of $1/3$. This is the value used in Midrigan's problem set 5.

Remark 1 (Damping convention). In the implementation, the *damping* parameter equals λ , the weight on the old guess. Some references (including Midrigan's notes) parametrize the update as:

$$K^{(n+1)} = K^{(n)} + s \left(\hat{K} - K^{(n)} \right)$$

where s is the "step size." The relationship is $s = 1 - \lambda$, so Midrigan's $s = 1/3$ corresponds to $\lambda = 2/3$.

2.2 The EGM Backward Step

Each backward step uses the Endogenous Grid Method of Carroll (2006). Given the continuation marginal value $V_{a,t+1}(a', z)$, the EGM step proceeds as follows:

1. Compute the expected marginal value of savings on the *exogenous* grid of end of period assets $a' \in \mathcal{A}$:

$$\mathbb{E}_t [V_{a,t+1}(a', z')] = \sum_{z'} \Pi(z' | z) V_{a,t+1}(a', z') \quad (12)$$

2. Invert the Euler equation to find consumption on this grid:

$$c_t^*(a', z) = (\beta(1 + r_t) \mathbb{E}_t [V_{a,t+1}(a', z')])^{-\sigma} \quad (13)$$

where σ is the coefficient of relative risk aversion (the inverse of the elasticity of intertemporal substitution σ^{-1}).

3. Use the budget constraint to recover the *endogenous* grid of beginning of period assets:

$$a_t^*(a', z) = \frac{c_t^*(a', z) + a' - y_t(z)}{1 + r_t} \quad (14)$$

4. Interpolate: the previous step produced pairs $(a_t^*(a'_k, z), a'_k)$ on an *endogenous* grid of beginning of period assets. To obtain policies on the *exogenous* grid $\mathcal{A} = \{a_1, \dots, a_N\}$ used by the rest of the algorithm, apply linear interpolation. For each $a_i \in \mathcal{A}$, find the bracketing pair $a_j^* \leq a_i < a_{j+1}^*$ and set

$$a'_t(a_i, z) = a'_j + \frac{a_i - a_j^*}{a_{j+1}^* - a_j^*} (a'_{j+1} - a'_j), \quad (15)$$

and analogously for consumption, $c_t(a_i, z) = c_j^* + \frac{a_i - a_j^*}{a_{j+1}^* - a_j^*} (c_{j+1}^* - c_j^*)$.

5. Enforce the borrowing constraint: if $a < a_t^*(\underline{a}, z)$, set $a'_t(a, z) = \underline{a}$ and $c_t(a, z) = (1 + r_t)a + y_t(z) - \underline{a}$.
6. Update the marginal value using the envelope condition:

$$V_{a,t}(a, z) = (1 + r_t) u'(c_t(a, z)) \quad (16)$$

2.3 The Forward Operator

Given a distribution D_t and a policy a'_t , the forward step computes D_{t+1} . Since policies generally do not land on grid points, we use lottery weights. For a given (a, z) with policy $a'_t(a, z)$, find the bracketing grid points $a_i \leq a'_t(a, z) < a_{i+1}$ and compute the weight:

$$\omega = \frac{a_{i+1} - a'_t(a, z)}{a_{i+1} - a_i} \quad (17)$$

Then distribute the mass $D_t(a, z) \cdot \Pi(z' | z)$ across (a_i, z') with weight ω and (a_{i+1}, z') with weight $1 - \omega$.

2.4 Resource Constraint Diagnostic

After convergence, we verify the resource constraint at each date:

$$Y_t - C_t - A_t + (1 - \delta)K_t = 0 \quad (18)$$

where $Y_t = Z_t K_t^\alpha$ is output, $C_t = \sum_{a,z} c_t(a, z) D_t(a, z)$ is aggregate consumption, and $A_t = \sum_{a,z} a'_t(a, z) D_t(a, z)$ is aggregate savings (next period's capital). The residual should be near machine precision ($\sim 10^{-10}$).

3 Method 2: Root Finding via Broyden's Method

An alternative to the shooting method is to cast the transition path as a system of non-linear equations and solve it directly using a quasi Newton method. This approach can converge faster (in fewer iterations), though each iteration is more expensive.

3.1 Formulation as a Root Finding Problem

Define the residual function $F : \mathbb{R}^T \rightarrow \mathbb{R}^T$ as follows. Given a candidate capital path $\mathbf{K} = (K_0, K_1, \dots, K_{T-1})$:

1. Compute prices $\{r_t, w_t\}$ from $\mathbf{K}$ and $\{Z_t\}$.
2. Run the backward iteration to obtain policies $\{a'_t, c_t\}$.
3. Run the forward iteration to obtain implied capital $\{\hat{K}_t\}$.
4. Return the residual:

$$F(\mathbf{K}) = \hat{\mathbf{K}} - \mathbf{K} \quad (19)$$

The equilibrium capital path satisfies $F(\mathbf{K}^*) = \mathbf{0}$. Steps 1 through 3 are identical to a single iteration of the shooting method.

3.2 Newton's Method and the Jacobian Problem

Newton's method would update the guess as:

$$\mathbf{K}^{(n+1)} = \mathbf{K}^{(n)} - \left[J_F(\mathbf{K}^{(n)}) \right]^{-1} F(\mathbf{K}^{(n)}) \quad (20)$$

where $J_F \in \mathbb{R}^{T \times T}$ is the Jacobian of F . Computing and inverting this Jacobian is prohibitively expensive for large T : each column requires a full backward forward pass (to compute one finite difference), so the cost is $O(T)$ evaluations of F , and inversion costs $O(T^3)$.

3.3 Broyden's Method: Rank 1 Jacobian Approximation

Broyden's method avoids computing the full Jacobian by maintaining a rank 1 approximation $B^{(n)} \approx J_F(\mathbf{K}^{(n)})$ that is updated at each step.

Algorithm 2 (Broyden's Method for Transition Paths).

Inputs: Initial guess $\mathbf{K}^{(0)}$, tolerance $\varepsilon > 0$.

Initialize: Set $B^{(0)} = I$ (identity matrix) or a finite difference approximation.

Step 1. Evaluate residual. Compute $F(\mathbf{K}^{(n)})$ via one backward forward pass.

Step 2. Convergence check. If $\|F(\mathbf{K}^{(n)})\|_\infty < \varepsilon$, stop.

Step 3. Compute search direction. Solve for the Newton step:

$$\Delta \mathbf{K}^{(n)} = - \left[B^{(n)} \right]^{-1} F(\mathbf{K}^{(n)}) \quad (21)$$

Step 4. Line search. Optionally perform a line search (Armijo condition) to find a step size α_n such that:

$$\|F(\mathbf{K}^{(n)} + \alpha_n \Delta \mathbf{K}^{(n)})\| < \|F(\mathbf{K}^{(n)})\|$$

Step 5. Update guess.

$$\mathbf{K}^{(n+1)} = \mathbf{K}^{(n)} + \alpha_n \Delta \mathbf{K}^{(n)}$$

Step 6. Update Jacobian approximation. Define:

$$\mathbf{s}^{(n)} = \mathbf{K}^{(n+1)} - \mathbf{K}^{(n)}$$

$$\mathbf{y}^{(n)} = F(\mathbf{K}^{(n+1)}) - F(\mathbf{K}^{(n)})$$

Apply the Broyden rank 1 update:

$$B^{(n+1)} = B^{(n)} + \frac{(\mathbf{y}^{(n)} - B^{(n)} \mathbf{s}^{(n)})(\mathbf{s}^{(n)})^\top}{\|\mathbf{s}^{(n)}\|^2} \quad (22)$$

Return to Step 1.

Why Broyden works well here. The key insight is that the Jacobian of the transition path residual F has a special structure. A change in K_t affects prices at date t , which propagates through the backward forward pass in a smooth, monotone fashion. The rank 1 updates capture this structure well after a few iterations, leading to superlinear convergence in practice.

3.4 Comparison of Methods

	Shooting	Root finding (Broyden)
Cost per iteration	1 backward forward pass	1 backward forward pass
Convergence rate	Linear	Superlinear
Typical iterations	25–40	10–20
Robustness	Very robust with damping	Sensitive to initial guess
Tuning parameter	Damping λ	Line search
Memory	$O(T)$	$O(T^2)$ (Jacobian approx.)

In practice, shooting is the default choice due to its robustness. Root finding is useful when speed matters (shorter T , good initial guess) or when damping is difficult to tune.

4 Method 3: Root Finding over Interest Rates

The previous section parameterized the root finding problem in terms of the capital path $\mathbf{K}$. An alternative, and more economically natural formulation, is to search over the interest rate path $\mathbf{r}$ and define the residual as the excess demand for capital: the difference between household capital supply and firm capital demand. This mirrors the steady state equilibrium condition $K^s(r) - K^d(r) = 0$ directly.

4.1 Capital Supply and Demand

In the Aiyagari model, the firm's first order condition pins down capital demand as a function of the interest rate and TFP:

$$K_t^d(r_t) = \left(\frac{Z_t \alpha}{r_t + \delta} \right)^{\frac{1}{1-\alpha}} \quad (23)$$

This is a static, period by period relationship: given r_t and Z_t , capital demand is uniquely determined.

Capital supply is the outcome of household optimization. Given an entire path of interest rates $\mathbf{r} = (r_0, \dots, r_{T-1})$, we can compute wages from K^d :

$$w_t = Z_t(1 - \alpha) \left(K_t^d \right)^\alpha \quad (24)$$

then run the backward forward iteration (using r_t and $y_t(z) = w_t z$ as inputs) to obtain the implied capital supply:

$$K_t^s(\mathbf{r}) = \sum_{a,z} a \cdot D_t(a, z) \quad (25)$$

Note that K_t^s depends on the *entire* path $\mathbf{r}$ (through the backward iteration), not just r_t .

4.2 Excess Demand Formulation

Define the excess demand function $F : \mathbb{R}^T \rightarrow \mathbb{R}^T$:

$$F(\mathbf{r}) = K^s(\mathbf{r}) - K^d(\mathbf{r}) \quad (26)$$

The equilibrium interest rate path satisfies $F(\mathbf{r}^*) = \mathbf{0}$: at every date, household asset supply equals firm capital demand.

Algorithm 3 (Broyden's Method over Interest Rates).

Inputs: Steady state interest rate $\bar{r}$, terminal marginal value V_a^{ss} , initial distribution D^{ss} , shock path $\{Z_t\}$, tolerance $\varepsilon > 0$.

Initialize: Set $r_t^{(0)} = \bar{r}$ for all t (flat at the steady state rate).

Step 1. Compute capital demand and wages. For each t :

$$K_t^d = \left(\frac{Z_t \alpha}{r_t^{(n)} + \delta} \right)^{\frac{1}{1-\alpha}}, \quad w_t = Z_t (1 - \alpha) (K_t^d)^\alpha$$

Step 2. Backward iteration. Using $\{r_t^{(n)}, w_t z\}$ as the price path, run EGM backward from V_a^{ss} to obtain policies $\{a'_t, c_t\}_{t=0}^{T-1}$.

Step 3. Forward iteration. Starting from $D_0 = D^{ss}$, iterate the distribution forward to obtain $K_t^s = \sum_{a,z} a \cdot D_t(a, z)$.

Step 4. Evaluate residual. $F(\mathbf{r}^{(n)}) = K^s - K^d$.

Step 5. Convergence check. If $\|F(\mathbf{r}^{(n)})\|_\infty < \varepsilon$, stop.

Step 6. Broyden update. Apply the rank 1 Jacobian update and compute the next guess:

$$\mathbf{r}^{(n+1)} = \mathbf{r}^{(n)} - \left[B^{(n)} \right]^{-1} F(\mathbf{r}^{(n)})$$

with $B^{(n+1)}$ updated via the same formula as in Algorithm 2. Return to Step 1.

4.3 Relationship to the Capital Path Formulation

The two formulations are equivalent in the following sense. The firm FOC (7) defines a bijection between r_t and K_t (for fixed Z_t, α, δ). At any point where $K_t = K^d(r_t)$, the residual vectors coincide:

$$K^s(\mathbf{r}) - K^d(\mathbf{r}) = \hat{\mathbf{K}}(\mathbf{K}) - \mathbf{K} \quad (27)$$

because K^s is computed from the same backward forward pass in both cases (using the same prices), and $K^d(r_t) = K_t$ by construction.

The formulations differ in their *search space*: one explores $\mathbb{R}^T$ in capital coordinates, the other in interest rate coordinates. This affects the Jacobian approximation maintained by Broyden's method and therefore the convergence path, even though the fixed point is identical. In practice:

- The $\mathbf{r}$ formulation is more natural economically: it directly generalizes the steady state bisection on r .
- The $\mathbf{K}$ formulation is closer to the shooting iteration: the residual $\hat{K} - K$ is the same quantity that the damped update drives to zero.
- Convergence speed may differ due to different conditioning of the Jacobian in the two coordinate systems.

5 Transition Dynamics as a Coordination Problem

The three methods above all solve the same equilibrium condition: aggregate capital supply equals aggregate capital demand at every date. But *why* does a solution exist, and why is it unique? These questions are not merely theoretical. The convergence properties of shooting and Broyden, the role of damping, and the conditioning differences between the $\mathbf{K}$ and $\mathbf{r}$ formulations all trace back to the structure of the equilibrium fixed point. This section develops the connection between the Aiyagari transition path and the framework of strategic interaction in macroeconomic models, following Angeletos and Lian (2016). The key insight is that the standard Aiyagari model exhibits stabilizing general equilibrium feedback, which guarantees a unique equilibrium and underlies the convergence of all three numerical methods.

5.1 The Aiyagari Economy as a Game

Even though households are atomistic (no single agent's savings decision moves prices), their collective behavior creates a feedback loop: individual savings determine aggregate capital, which determines prices, which in turn shape individual savings incentives. This feedback has the structure of a game among a continuum of players. Making this structure explicit connects the computational problem to the theory of coordination in macroeconomics.

Definition 1 (Bewley–Aiyagari game). *The Aiyagari economy defines the following game:*

- **Players:** a continuum of households $i \in [0, 1]$.

- **Individual state (“fundamental”):** $\theta_i = (a_i, z_i) \in \mathcal{A} \times \mathcal{Z}$, the asset and productivity pair.
- **Action:** $k'_i \in [\underline{a}, \bar{a}]$, the savings choice (next period capital holding).
- **Aggregate outcome:** $K = \int k'_i dF(\theta_i)$, the mean savings across the population.
- **Payoff:** The lifetime utility $V_t(a_i, z_i; \{r_s, w_s\}_{s \geq t})$ from the Bellman equation (1), where prices depend on aggregate capital through the firm first order conditions (7)–(8).

Under the MIT shock assumption, every household has perfect foresight over the entire future path of the economy. In the terminology of Angeletos and Lian (2016), information is *perfect*: no agent faces fundamental uncertainty (each knows θ_i and the cross-sectional distribution Θ) or strategic uncertainty (each can compute the equilibrium K from common knowledge of the state). Higher order beliefs coincide with first order beliefs, and coordination is perfect.

Remark 2 (Atomistic agents and the “game” interpretation). *The game-theoretic language does not imply that agents reason strategically. Each household treats prices as exogenous and optimizes individually; the “strategic” interaction operates entirely through general equilibrium. The rational expectations equilibrium imposes that all agents perceive the same price path, and that this commonly perceived path coincides with the one generated by the joint behavior of all agents. This is the standard consistency requirement of a competitive equilibrium, recast in the language of a symmetric Bayesian Nash equilibrium following Angeletos and Lian (2016, Definition 2).*

5.2 The Aggregate Fixed-Point Equation

The equilibrium of the Aiyagari economy can be characterized as a fixed point: the capital path agents collectively choose, given the prices implied by that path, must reproduce the path itself. We now define this mapping precisely and connect it to the residual functions from the computational methods.

Definition 2 (Implied capital mapping). *Define $G : \mathbb{R}^T \rightarrow \mathbb{R}^T$ as follows. Given a candidate capital path $\mathbf{K} = (K_0, \dots, K_{T-1})$:*

1. *Compute prices: $r_t = Z_t \alpha K_t^{\alpha-1} - \delta$, $w_t = Z_t(1 - \alpha)K_t^\alpha$.*
2. *Run the backward iteration from V_a^{ss} to obtain policies $\{a'_t\}_{t=0}^{T-1}$.*
3. *Run the forward iteration from $D_0 = D^{ss}$ to obtain implied capital $\hat{K}_t = \sum_{a,z} a \cdot D_t(a, z)$.*

Then $G(\mathbf{K})_t = \hat{K}_t$ is the implied capital at each date.

The equilibrium condition is the fixed point

$$\mathbf{K} = G(\mathbf{K}), \tag{28}$$

which states that the capital path guessed by agents equals the capital path produced by their collectively optimal behavior at the corresponding prices. This is the T -dimensional analogue of the scalar fixed point equation $K = G(K, \mathbf{B})$ in Angeletos and Lian (2016, Proposition 3), where $\mathbf{B}$ is the cross-sectional distribution of beliefs about fundamentals. Under perfect information, $\mathbf{B}$ is degenerate at the true distribution, and the fixed point reduces to (28).

The residual functions from the three computational methods are all derived from G :

$$F(\mathbf{K}) = G(\mathbf{K}) - \mathbf{K} \quad (\text{Method 2, root finding over } \mathbf{K}) \quad (29)$$

$$F(\mathbf{r}) = K^s(\mathbf{r}) - K^d(\mathbf{r}) \quad (\text{Method 3, root finding over } \mathbf{r}) \quad (30)$$

The shooting iteration (Method 1) is the damped fixed point iteration $\mathbf{K}^{(n+1)} = \lambda \mathbf{K}^{(n)} + (1 - \lambda) G(\mathbf{K}^{(n)})$. All three methods search for the same object: a zero of $G(\mathbf{K}) - \mathbf{K}$.

5.3 Strategic Interaction and Equilibrium Uniqueness

Whether the fixed point (28) is unique, and whether iterative methods converge to it, depends on the *slope* of G : how strongly the implied capital responds to a perturbation in the guessed capital. This slope encodes the nature of strategic interaction in the economy.

Steady state analysis. Consider first the scalar (steady state) case, where the fixed point is $K = G(K)$ with $G(K) = K^s(r(K), w(K))$. The derivative G_K measures the general equilibrium feedback: if every household saves one unit more, how much does the implied aggregate capital change?

Differentiating through the price functions:

$$G_K = \underbrace{\frac{\partial K^s}{\partial r} \cdot \frac{dr}{dK}}_{\text{interest rate channel}} + \underbrace{\frac{\partial K^s}{\partial w} \cdot \frac{dw}{dK}}_{\text{wage channel}} \quad (31)$$

The two channels capture two distinct mechanisms through which aggregate capital feeds back into individual savings:

1. **Interest rate channel** (stabilizing). More aggregate capital lowers the marginal product of capital: $dr/dK = \alpha(\alpha - 1)ZK^{\alpha-2} < 0$. A lower interest rate reduces the return on savings. For standard calibrations, the net effect on aggregate savings is negative ($\partial K^s/\partial r > 0$ combined with $dr/dK < 0$), so this channel makes each household want to save *less* when others save more. In the language of Angeletos and Lian (2016), this is *strategic substitutability*: the general equilibrium adjustment attenuates the partial equilibrium response.
2. **Wage channel** (destabilizing). More aggregate capital raises the marginal product of labor: $dw/dK = \alpha(1 - \alpha)ZK^{\alpha-1} > 0$. Higher wages increase labor income, which raises aggregate savings ($\partial K^s/\partial w > 0$). This channel makes each household want to save *more* when others save more: *strategic complementarity*. The GE adjustment amplifies the PE response.

The sign and magnitude of G_K depend on which channel dominates. In the standard Aiyagari calibration (Cobb-Douglas production, no externalities, no aggregate demand spillovers), the interest rate channel is the dominant force. However, the strength of this channel is significantly affected by the precautionary savings motive, as we now discuss.

Remark 3 (Precautionary savings and the interest rate channel). *The interest rate channel operates through $\partial K^s / \partial r$, the response of aggregate savings to changes in the return on assets. In a representative agent economy, the Euler equation pins down $r = 1/\beta - 1$ in steady state, and any deviation from this rate induces a large savings response: $\partial K^s / \partial r$ is effectively infinite, making the interest rate channel maximally strong.*

In the Bewley-Aiyagari economy, idiosyncratic risk and borrowing constraints introduce three forces that dampen $\partial K^s / \partial r$:

1. *Constrained agents do not respond. A positive measure of households sits at the borrowing constraint, $a'_i = \underline{a}$, regardless of r . For these agents $\partial k'_i / \partial r = 0$, contributing zero to the aggregate elasticity.*
2. *The buffer stock target is inelastic. Agents near the constraint maintain a precautionary buffer whose size is governed primarily by the variance of income shocks and the tightness of the borrowing limit, not by the return on savings. This buffer responds weakly to r .*
3. *Precautionary income effect. A higher r means the existing precautionary buffer earns a higher return, so agents can maintain the same level of self-insurance with less saving. This is an income effect that opposes the substitution effect. Under CRRA utility with $\gamma > 1$, the income effect is amplified by the precautionary motive: the marginal value of the buffer falls when the buffer grows faster, reducing the incentive to add to it.*

The net effect on $\partial K^s / \partial r$ depends on the calibration. For moderate risk aversion ($\gamma \leq 2$), the substitution effect typically dominates and $\partial K^s / \partial r > 0$, preserving the stabilizing nature of the interest rate channel, though with reduced magnitude. For high risk aversion ($\gamma \geq 3$) combined with high income volatility, the precautionary income effect can dominate, pushing $\partial K^s / \partial r$ toward zero or even making it negative (Aiyagari, 1994, Table III).

When $\partial K^s / \partial r < 0$, the interest rate channel flips sign: $(\partial K^s / \partial r)(dr/dK) = (-) \times (-) > 0$. Instead of strategic substitutability, the interest rate channel now contributes to strategic complementarity: more aggregate capital lowers r , which through the dominant income effect raises savings further. In this regime, both channels push G_K in the positive direction, making complementarity more likely. The standard Aiyagari calibration (Section 6) has $\gamma = 1.5$, which lies in the substitution-dominated regime, so the interest rate channel is stabilizing. But the general lesson is that incomplete markets weaken the interest rate channel relative to the representative agent benchmark, thereby shifting the economy in the direction of strategic complementarity.

Definition 3 (Strategic interaction classification). *Following Angeletos and Lian (2016, Definitions 5–6):*

- *The economy exhibits strategic substitutability if $G_K < 0$: the GE feedback reverses the direction of the PE effect.*
- *The economy exhibits weak strategic complementarity if $G_K \in (0, 1)$: the GE feedback amplifies the PE effect, but by less than one for one.*
- *The economy exhibits strong strategic complementarity if $G_K > 1$ at some equilibrium: the GE feedback is self-reinforcing, and multiple equilibria can arise.*

Proposition 1 (Equilibrium uniqueness). *Consider the Aiyagari model with Cobb-Douglas production, no externalities, and standard parameter restrictions ($\alpha \in (0, 1)$, $\delta > 0$, $\beta < 1$, $r^* < 1/\beta - 1$). If $|G_K| < 1$ at every fixed point $K = G(K)$, then:*

- (i) *The steady state equilibrium exists and is unique.*

- (ii) The undamped fixed point iteration $K^{(n+1)} = G(K^{(n)})$ converges to K^* from any initial guess in a neighborhood of K^* .

Proof. Existence follows from the intermediate value theorem: G maps a compact interval $[a, \bar{a}]$ continuously into itself (the asset grid is bounded and the distribution has finite mean). For uniqueness: under strategic substitutability ($G_K < 0$), $G(K) - K$ is strictly decreasing, so it crosses zero at most once. Under weak complementarity ($G_K \in (0, 1)$), G is a contraction; the Banach fixed point theorem gives both uniqueness and convergence of the iteration. In either case, exactly one fixed point exists. Part (ii) follows from the contraction mapping theorem when $G_K \in [0, 1)$, and from monotone convergence of $G(K) - K$ when $G_K < 0$ and $|G_K| < 1$. $\square$

Proposition 1 connects the economic structure of the model to the existence of a well-posed computational problem. The condition $|G_K| < 1$, which rules out strong complementarity, is what makes the transition path problem solvable by iterative methods. The same condition, applied to the T -dimensional Jacobian ∇G , governs convergence of the dynamic methods (Section 5.5).

Remark 4 (What would break uniqueness). *The standard Aiyagari model lacks mechanisms that would push G_K above 1. In the neoclassical growth model example of Angeletos and Lian (2016), adding Romer-style production spillovers $Y = ZK^\alpha L^{1-\alpha} \cdot K^\gamma$ with γ sufficiently large creates strong complementarity: more aggregate capital directly raises each firm's productivity, generating a positive feedback that can sustain multiple equilibria. In the heterogeneous agent literature, mechanisms that could generate strong complementarity include financial accelerators (where net worth affects borrowing capacity, creating a positive feedback between aggregate capital and individual investment), aggregate demand externalities, and search frictions with thick market effects. None of these are present in the standard Bewley-Aiyagari framework, which is why the equilibrium is unique and the computational methods converge.*

Remark 5 (Connection to the excess demand formulation). *The steady state equilibrium condition can equivalently be written as the excess demand for capital:*

$$K^s(r) - K^d(r) = 0, \quad (32)$$

which is the scalar version of the $\mathbf{r}$ -formulation (26). The condition $|G_K| < 1$ translates into a slope condition on the excess demand: at the equilibrium interest rate r^ , the excess demand function $K^s(r) - K^d(r)$ crosses zero with a well-defined sign. In the Aiyagari model, $K^d(r)$ is strictly decreasing (from $+\infty$ as $r \rightarrow -\delta^+$ to 0 as $r \rightarrow +\infty$) and $K^s(r)$ is bounded and increasing for $r < 1/\beta - 1$, so the excess demand crosses zero exactly once: the same uniqueness result, seen from the r -side.*

5.4 Computational Implications

The derivative G_K is not merely a theoretical quantity. It directly governs the convergence behavior of each computational method, providing a unified explanation for the numerical phenomena observed in practice.

Damping in the shooting method. The damped iteration (11) has the effective derivative

$$\frac{d}{dK} [\lambda K + (1 - \lambda)G(K)] = \lambda + (1 - \lambda)G_K. \quad (33)$$

Convergence requires $|\lambda + (1 - \lambda)G_K| < 1$. Three cases arise:

- *Weak complementarity* ($G_K \in (0, 1)$): the undamped iteration ($\lambda = 0$) already converges. Damping slows convergence but is not needed for stability.
- *Substitutability with* $|G_K| > 1$: the undamped iteration oscillates and diverges. Damping stabilizes the oscillations by mixing old and new guesses, reducing the effective slope below 1 in magnitude.
- *Strong complementarity* ($G_K > 1$): no value of $\lambda \in [0, 1)$ makes the effective slope less than 1 in magnitude. The fixed point is unstable under any damped iteration. This is the computational signature of multiple equilibria.

The fact that shooting converges with $\lambda = 2/3$ (Midrigan's step size $s = 1/3$) in the Aiyagari model confirms that the spectral radius of the GE feedback is bounded, consistent with unique equilibrium.

Broyden convergence and the Jacobian. Broyden's method approximates the Jacobian of $F(\mathbf{K}) = G(\mathbf{K}) - \mathbf{K}$. In the scalar case, this Jacobian is $G_K - 1$:

- When G_K is far from 1 (substitutability or weak complementarity), $|G_K - 1|$ is well separated from zero. The Jacobian is well conditioned, and Broyden converges rapidly.
- When $G_K \approx 1$ (near the boundary of strong complementarity), $G_K - 1 \approx 0$. The Jacobian is nearly singular: Newton-type methods slow down or fail. This is the computational signature of near-indeterminacy.

The rapid convergence of Broyden in the standard Aiyagari model (~ 15 iterations) confirms that G_K is well below 1, placing the economy firmly in the determinate regime.

Conditioning in K vs. r coordinates. The two root finding formulations solve the same equilibrium but parameterize the search differently. Their Jacobians are related by the chain rule. In the scalar case:

$$F'_K(K) = G_K - 1 \quad (\text{Jacobian in } K \text{ coordinates}) \quad (34)$$

$$F'_r(r) = \frac{dK^s}{dr} - \frac{dK^d}{dr} \quad (\text{Jacobian in } r \text{ coordinates}) \quad (35)$$

Since the firm FOC defines a bijection $K^d(r(K)) = K$, differentiating gives $dK^d/dr \cdot dr/dK = 1$, so the Jacobians satisfy

$$F'_K(K) = F'_r(r) \cdot \frac{dr}{dK}. \quad (36)$$

The factor $dr/dK = \alpha(\alpha - 1)ZK^{\alpha-2}$ is small in magnitude for typical calibrations (of order 10^{-4} , since K is large and $\alpha - 1 < 0$). This means $|dK/dr|$ is large: small changes

in r correspond to large changes in K . Equivalently, the r -formulation compresses a wide range of capital variation into a narrow range of interest rates.

For Broyden's method, this compression is problematic when the initial Jacobian approximation is $B^{(0)} = I$. The identity is a reasonable approximation in K coordinates (the residual and the variable are both in capital units), but a poor one in r coordinates (the residual is in capital units while the variable is in interest rate units, with a scale mismatch of order $|dK/dr| \approx 3,654$). This explains why the r -formulation requires rescaling for efficient convergence.

The rescaled variable $x_t = (r_t - \bar{r}) \cdot s$ with

$$s = \frac{K^*}{(\bar{r} + \delta)(1 - \alpha)} \approx \left| \frac{dK}{dr} \right|_{r=\bar{r}} \quad (37)$$

transforms the Jacobian to $F'_x = F'_r/s \approx F'_K$, restoring the natural conditioning of the K -formulation and enabling convergence from a unit initialization.

5.5 The Dynamic Generalization

For the transition path, the scalar analysis extends to T dimensions. The implied capital mapping $G : \mathbb{R}^T \rightarrow \mathbb{R}^T$ (Definition 2) has a Jacobian $\nabla_K G \in \mathbb{R}^{T \times T}$ with entries

$$[\nabla G]_{t,s} = \frac{\partial \hat{K}_t}{\partial K_s}, \quad (38)$$

which encode how a perturbation to capital at date s (through the prices it generates) propagates to the implied capital supply at date t . This matrix has a rich structure:

- *Diagonal entries* $[\nabla G]_{t,t}$: the within-period GE feedback, analogous to the scalar G_K . A change in K_t alters prices at date t , which directly affects savings decisions at t .
- *Off-diagonal entries* $[\nabla G]_{t,s}$ for $s \neq t$: the intertemporal GE feedback. A price change at date s propagates backward through the Euler equation (affecting policies at dates $t < s$) and forward through the distribution dynamics (affecting capital supply at dates $t > s$).

The equilibrium is unique and the damped shooting iteration converges whenever the spectral radius satisfies

$$\rho(\nabla G) < 1, \quad (39)$$

which is the T -dimensional generalization of $|G_K| < 1$. With damping λ , the effective iteration matrix is $(1 - \lambda)\nabla G + \lambda I$, and convergence requires $\rho((1 - \lambda)\nabla G + \lambda I) < 1$. This is satisfied for sufficiently large λ whenever $\rho(\nabla G)$ is finite, which holds in the Aiyagari model since the GE feedback is bounded.

The off-diagonal entries of ∇G decay with $|t - s|$: a perturbation at a distant date has an attenuated effect on current capital supply, because the backward Euler equation and forward distribution dynamics are both contractive. This gives the Jacobian a nearly banded structure that Broyden's rank 1 updates approximate well, explaining the rapid convergence (~ 15 iterations for $T = 300$) observed in practice.

Remark 6 (Summary: economic structure and numerical performance). *The connections developed in this section are summarized in the following table.*

<i>Economic property</i>	<i>Formal condition</i>	<i>Computational consequence</i>
<i>Diminishing returns</i>	$dr/dK < 0$	<i>Stabilizing GE feedback</i>
<i>Substitutability / weak compl.</i>	$ G_K < 1$	<i>Unique equilibrium</i>
<i>Bounded spectral radius</i>	$\rho(\nabla G) < 1$	<i>Shooting converges (with damping)</i>
<i>Well-conditioned Jacobian</i>	G_K far from 1	<i>Broyden converges in ~ 15 iters</i>
<i>$K \leftrightarrow r$ bijection</i>	$ dK/dr \gg 1$	<i>Rescaling needed for r-formulation</i>
<i>Decaying off-diagonals</i>	$ [\nabla G]_{t,s} \rightarrow 0$ as $ t-s \rightarrow \infty$	<i>Rank 1 updates are effective</i>

6 Application: TFP Shock in the Aiyagari Model

Consider the standard Aiyagari (1994) economy with a Cobb Douglas production function $Y_t = Z_t K_t^\alpha$. At $t = 1$, an unexpected TFP shock hits:

$$\begin{aligned} \log Z_0 &= 0, & \log Z_1 &= 0.10 \\ \log Z_t &= 0.95 \log Z_{t-1}, & t &\geq 2 \end{aligned} \tag{40}$$

The price function that maps aggregate capital and time to household inputs is:

$$\begin{aligned} r_t &= Z_t \alpha K_t^{\alpha-1} - \delta \\ w_t &= Z_t (1 - \alpha) K_t^\alpha \\ y_t(z) &= w_t \cdot z \end{aligned} \tag{41}$$

Calibration. The baseline parameters are: $\beta = 0.99$, $\gamma = 1.5$ (CRRA), $\rho_z = 0.99$, $\sigma_\varepsilon = 0.10$, $\alpha = 0.35$, $\delta = 0.02$, with 11 Rouwenhorst income states and 501 asset grid points up to $\bar{a} = 2500$.

Results. The shooting method with $\lambda = 2/3$ converges in approximately 34 iterations. Capital rises above the steady state (agents save more in response to higher TFP and wages), then decays back as the shock fades. The resource constraint residual is near machine precision ($\sim 10^{-10}$). The root finding method (Broyden) matches the shooting solution to within $\sim 10^{-8}$ in capital.

7 Implementation Notes

The solver in `hetmacro.transition` is model agnostic. The user provides:

1. Steady state dictionaries (from `household_steady_state`) containing the distribution D , marginal value V_a , policies, and aggregates.
2. A price function `price_fn(K, t) -> dict` that maps aggregate capital and time to household inputs $\{r, y, w\}$.

Key design choices.

- The backward step defaults to EGM (`backward_egm`), but a custom `backward_fn` can be provided to use any solver (VFI, Howard improvement, Euler iteration) via the `make_backward_fn` wrapper.
- Capital at time t is computed as $K_t = \sum_{a,z} a \cdot D_t(a, z)$ (mean assets in the distribution), not as aggregate savings from policies. This avoids a timing mismatch between the guess and the implied path.
- The income argument to the steady state solver must be $y = w \cdot z$ (income levels), not raw productivity z . This ensures consistency between the steady state and the transition price function.

References

- [1] S. R. Aiyagari. Uninsured idiosyncratic risk and aggregate saving. *Quarterly Journal of Economics*, 109(3):659–684, 1994.
- [2] G.-M. Angeletos and C. Lian. Incomplete information in macroeconomics: Accommodating frictions in coordination. In J. B. Taylor and H. Uhlig, editors, *Handbook of Macroeconomics*, volume 2A, chapter 9, pages 1065–1240. Elsevier, 2016.
- [3] C. G. Broyden. A class of methods for solving nonlinear simultaneous equations. *Mathematics of Computation*, 19(92):577–593, 1965.
- [4] C. D. Carroll. The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics Letters*, 91(3):312–320, 2006.
- [5] R. Cooper and A. John. Coordinating coordination failures in Keynesian models. *Quarterly Journal of Economics*, 103(3):441–463, 1988.
- [6] P. A. Diamond. Aggregate demand management in search equilibrium. *Journal of Political Economy*, 90(5):881–894, 1982.
- [7] P. M. Romer. Increasing returns and long-run growth. *Journal of Political Economy*, 94(5):1002–1037, 1986.